STUDY | Research Results, Data Analysis and Scientific Visualization

Research: Trust in AI-Generated Medical Recommendations Among Patients: Three Age Groups

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GitHub: github.com/marzhanolzhabek/ai-medical-trust-research

1. Research design and data provenance

Cross-sectional survey demonstration comparing ages 18–30, 31–50, and 51+. The sample includes 45 **synthetic** records (15 per group), not actual patient responses. Trust is measured on an ordinal five-point scale; survey group means/intervals are descriptive examples only and cannot establish real clinical findings or confirm hypotheses.

Age group	n	Mean trust	Median	SD	95% CI of mean
18-30	15	3.87	4	0.83	3.47–4.27
31-50	15	3.20	3	1.01	2.73–3.67
51+	15	3.07	3	0.80	2.67–3.47

2. Controlled software-processing benchmark

Baseline: three full scans to calculate cohort counts/means. **Proposed:** one-pass validation/aggregation, generating identical outputs. Four workload sizes (45, 90, 180, 360) repeat the same 45 synthetic responses with technical IDs. Repeats are computational inputs, not additional patients. Timing covers in-memory aggregation only, excluding CSV I/O, bootstrap survey CIs, and any medical-AI model. Measurements were collected in a separate execution environment, not on the students' own computer.

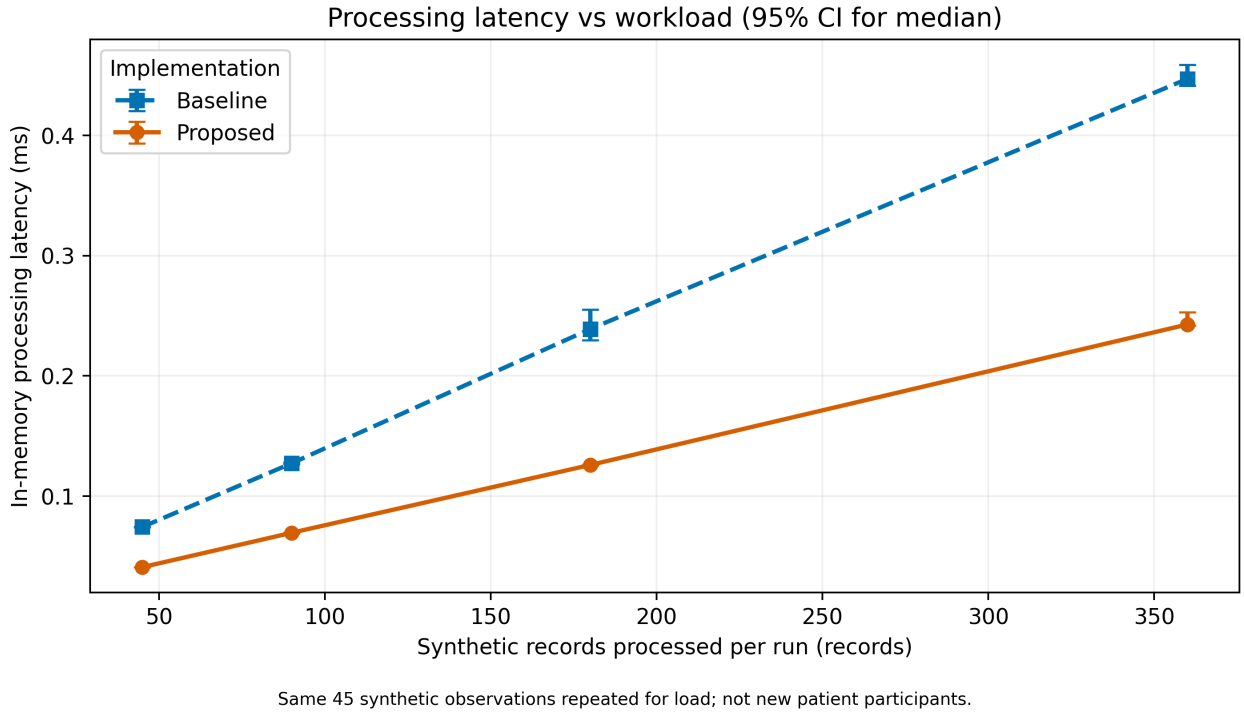
For each implementation and load: ten labeled warm-up executions excluded from statistics, then eighty benchmark executions. Total: 720 JSONL rows; 640 measured executions. Error rate is computed over all evaluation runs; p25/p75, median, p95, p99 and bootstrapped 95% confidence limits are computed over successful measurements. The bootstrap uses 2,000 repetitions and seed 42.

Records	Method	Median (ms)	p25–p75 (ms)	p95 / p99 (ms)	95% CI median (ms)
45	Baseline	0.0744	0.0738–0.0753	0.1371 / 0.2075	0.0740–0.0747
45	Proposed	0.0409	0.0407–0.0413	0.0542 / 0.0737	0.0409–0.0410
90	Baseline	0.1273	0.1263–0.1315	0.1889 / 0.2612	0.1268–0.1280
90	Proposed	0.0695	0.0689–0.0702	0.0990 / 2.3808	0.0692–0.0698
180	Baseline	0.2390	0.2270–0.3298	0.4921 / 1.0940	0.2298–0.2554
180	Proposed	0.1259	0.1248–0.1333	0.2065 / 0.3574	0.1257–0.1264
360	Baseline	0.4471	0.4353–0.4716	0.5955 / 0.7881	0.4416–0.4589
360	Proposed	0.2428	0.2397–0.2618	0.3186 / 0.4013	0.2419–0.2532

Measured error rate: 0% across all eight method/load groups. These machine-specific latency measurements show an improvement for the proposed one-pass method on this synthetic repeated-input workload, not a general clinical or production-performance guarantee.

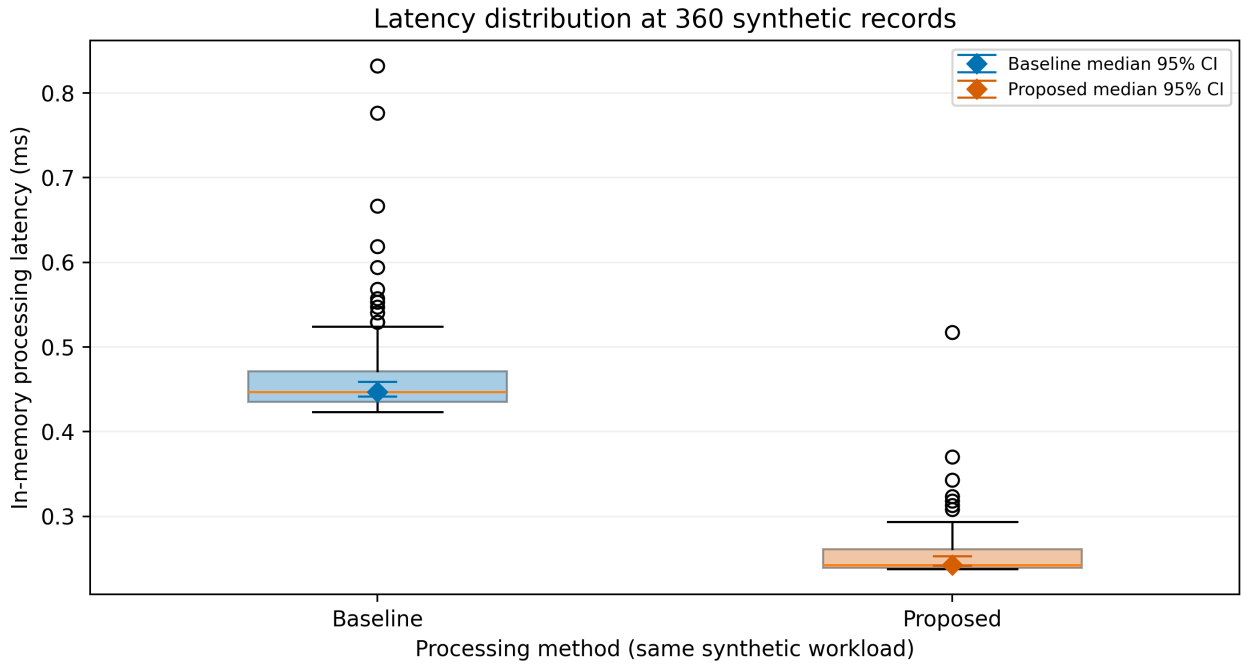
3. Software processing plots

Scientific plots use explicit units, colorblind-readable styles and bootstrap 95% confidence intervals for the median. Vector PDF and 350 DPI PNG versions are also provided inside the repository.



Same 45 synthetic observations repeated for load; not new patient participants.

Figure 1. Measured latency against increasing repeated synthetic-record workload. Bars = 95% bootstrap CI for median.



Boxes: IQR and whiskers; diamonds: medians with 95% bootstrap CI.

Figure 2. Benchmark latency distributions at 360 records; diamonds with error bars = median and 95% CI.

4. Synthetic survey visualizations

These graphics are separate from processing performance. All values are generated synthetic examples; no real survey data have been collected.

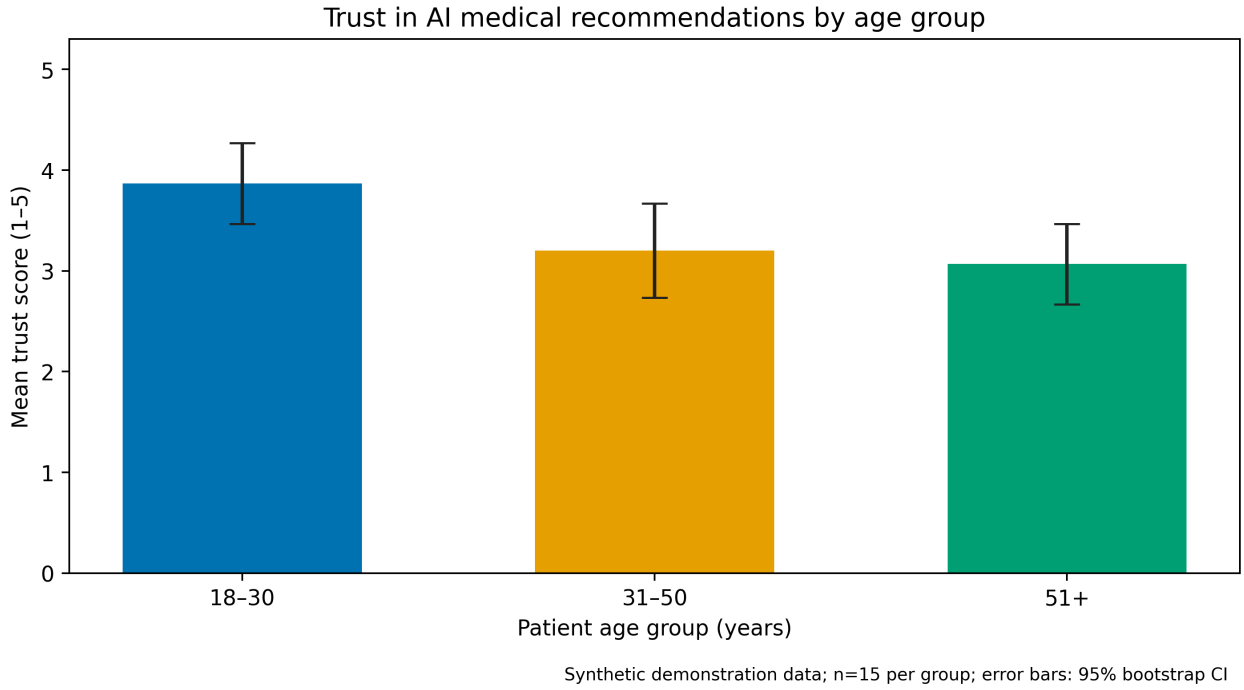


Figure 3. Synthetic trust means across age categories, with 95% bootstrap CI for mean.

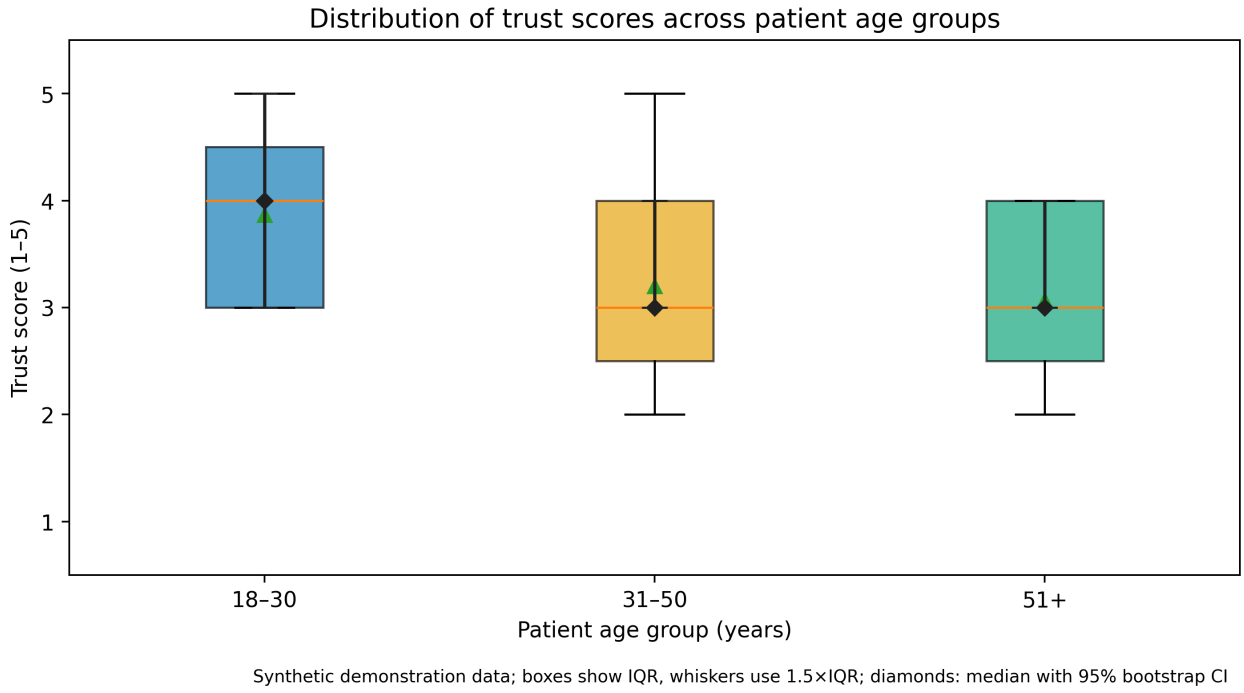


Figure 4. Synthetic trust score distribution, with 95% bootstrap confidence bars around group medians.

5. C4 domain model, reproducibility and conclusions

The C4 component diagram in **docs/domain_model.md** identifies two distinct metric collection points: (M1) survey trust scores and summary intervals; (M2) JSONL software timing, percentile latency and error-rate guardrail. Python processing and visualization are separated into scripts. The Jupyter notebook executes sequentially with stored outputs and no external downloads.

To reproduce from the repository root: install **requirements.txt**; run **python src/compare_methods.py**, **python src/benchmark_statistics.py**, **python src/benchmark_visualize.py** and **python -m unittest discover -s src -p "test_*.py"**. Recollecting timings changes the numerical results with local computer conditions. Notebook execution uses pre-recorded JSONL and does not regenerate it.

Conclusion: the one-pass implementation decreases processing latency relative to repeated scans in this controlled synthetic benchmark. The trust figures are demonstrations, not evidence of actual population patterns. Before real-world patient interpretation, the study needs ethical approval as appropriate, informed consent, data collection and further validation.